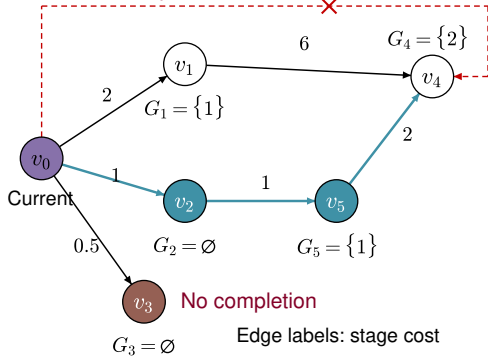


I. Motion vs. completion

Illustrative graph · budget $h = 3$

$s = (0, 0, 0, 3)$ $\mathcal{K} = \{1, 2\}$

Rejected transfer $\chi_{04} = 0$



Zero-gain views can be useful connectors.

II. Cost-to-go changes the decision

$$\widehat{Q}_d(s, a) = \ell_{ja} + \widehat{V}_{d-1}(f(s, a))$$

Exact prefix $d = 2$ + full base-policy tail

$a \in \mathcal{U}_s(s)$

Action a	ℓ_{0a}	$\widehat{V}_1(f(s, a))$	$\widehat{Q}_2(s, a)$	Decision
1	2	6	8	Base policy
2	1	$1 + 2 = 3$	4	Selected
3	0.5	$+\infty$	$+\infty$	Discard

Rollout selects the lowest finite total cost.

III. Replan with a retained completion

Take $a^* = 2$

$s' = (2, 0, \mathbf{e}_2, 2)$



$$J_{\text{rollout}}(s) = 4 = \widehat{V}_2(s) < \widehat{V}_0(s) = 8$$